

Deep Learning Approach for Prediction Employee Attrition for HR Analytics

T.E.Ramya¹

Department of Computer Science and Engineering
Kongu Engineering College
Erode, India
ramyaerusan@gmail.com

B.Sanjay²

Department of Computer Technology
Kongu Engineering College
Erode, India
sanjaybelli28@gmail.com

Abstract—Employee attrition has been a major challenge for organizations as it influences the productivity and the continuity of operations within any organization. The objective of this paper is to identify some of the crucial factors that lead to attrition through the analysis of employee data by providing an effective model that can be used by HR for decision-making. This paper focuses on an analysis of employee attrition where traditional ML techniques are used in tandem with advanced deep learning models for predicting and understanding the factors leading to employee turnover. In this paper, traditional ML models such as Logistic Regression and Random Forest Classifier, while being used in combination with a voting classifier ensemble and an integrated hybrid deep learning model based on CNNs and LSTM networks. Scaling of numerical features and one-hot encoding for the categorical variables will be implemented to ensure that the data is appropriately structured for each model to apply 'HR-Employee-Attrition.csv'. The Voting Classifier leverages predictions from multiple machine learning models to increase the possibility of providing an accurate answer. The deep learning model, using CNN layers but further enhanced with LSTM layers, can identify structural patterns as well as further temporal patterns within employee data. Using the models, a performance metric of accuracy score was 0.87, ROC-AUC was 0.54, along with confusion matrices to compare the deep learning model with other models for a better performance in terms of predictive power.

Keywords—Employee Attrition, Deep learning, Voting classifier, LSTM, CNN, accuracy, precision, recall, f1 score.

I. INTRODUCTION

This area of research into employee attrition has been a hot topic in the past few years, primarily because of its potential influence on the stability and productivity of an organization. With precise predictions concerning the likelihood of employees leaving an organization, businesses can resort to retention strategies proactively and, therefore, be able to better plan a workforce and save money on turnover costs. Classical methods most used in the approach include logistic regression, random forests, and NB classifiers[9]. This paper lays down a comparison of hybrid deep learning and traditional machine learning ensemble methods applied to the well-known dataset[7][8][13]. The hybrid deep learning model integrates CNN, and LSTM [1][2][10]-[12]. In contrast, the traditional ensemble technique depends on a Voting Classifier that aggregates the predictions from three different classifiers: Logistic Regression, Random Forest, and Gaussian Naive Bayes to improve the prediction performance[8][13]. This paper is meant to examine and compare the performances of these two approaches on the problem of predicting employee attrition, identifying whether deep learning methodologies are

providing any significant performance improvement above the corresponding ones found in traditional machine learning models in this domain of structured data [1][2][11][12]. Applied evaluation metrics are accuracy(0.87), ROC-AUC scores(0.54) and confusion matrices. Experimental results reveal that the discriminative capability of the hybrid DL model is slightly higher than the traditional ensemble approach. Employee attrition is a serious issue for any organization, as it involves the following: Knowledge Loss: Outgoing experienced employees take away the institutional knowledge and expertise with them. Recruitment Costs: Organisations have to pay high recruitment costs and train new recruits. Productivity Impact: Ripple effects of the loss of key people on team dynamics and timelines. Morale Effects: Attrition hurts the morale and engagement levels of the remaining employees. Advancements in machine learning techniques, especially deep learning, can improve predictability and intervention in cases of employee attrition[1][2][11]. The use of a combination of LSTM networks that identify temporal patterns and CNNs that extract features helps an organization anticipate and prepare better for possible risks of employee turnover[7][10].

II. LITERATURE REVIEW

Employee attrition prediction has received substantial interest in recent years as a result of its significance in workforce management and organizational efficiency. Many studies have explored the application of machine learning (ML) techniques and data analytics for predicting employee turnover and understanding its underlying factors. Angrave et al. (2016) emphasized the challenges HR departments face in using big data effectively, calling for strategic alignment between HR practices and analytics. Building on these ground, Punnoose, 2016 used have researched the use of machine learning (ML) techniques and made evident drivers of attrition. Few recent works have focused upon the tuning of prediction models by adopting sophisticated methods. Alharbi et al., 2024 and Bhartiya et al., 2019 reported that decision tree-based classification models like SVM which might provide the actionable knowledge in terms of the direction of attrition trends. Mansor et al. (2021) and Lhaksmana et al. (2022) pointed out feature selection techniques, like information gain and random forest classification, to enhance the accuracy and interpretability of a model. Lazzari et al. (2022) and Colomo-Palacios et al. (2021) employed deep data analytics and explainable AI to provide transparent predictions in the explanatory dimensions of attrition, which calls for a comprehension of causal relations in attrition to support the making of decisions. Fallucchi et al. (2020) established the applicability of ML techniques, such as logistic regression and neural networks, for turnover prediction. Jayakrishnan et al. (2022), Kadhem et al. (2022), and Alduayj et al. (2018)

contributed to the fact that comparative analysis of ML algorithms is important. Their studies showed that ensemble approaches, including random forests and boosting, are better suited for handling complex attrition datasets. Overall, the literature underlines the increasing applicability of ML in predicting employee attrition. These are the robust preprocessing of data, feature engineering, and explainable algorithms that bring actionable insights to the organization. That, in turn, allows them to do proactive retention.

III. PROPOSED METHODOLOGY

This proposed system of predicting attrition for employees integrates the deep learning model with the traditional machine learning model to give a full analysis of the factors resulting in attrition[1][2][11]. This hybrid system of integrating a CNN model along with LSTM layers and the voting classifier of traditional models ensures an effective capture of data. The hybrid model employs CNN and LSTM layers. CNN captures data patterns, and LSTM layers identify temporal dependencies[10]. They both improve the model's capabilities toward the recognition of time-based patterns and the figure1 shows the Structure of Proposed Model.

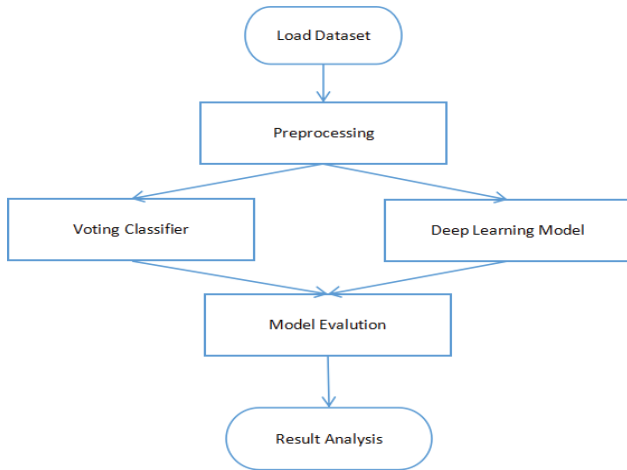


Fig. 1. Overall Structure of Proposed Model

Related to attrition. Data is used in training (80%) and testing (20%) sets for better generalization and preventing overfitting. Standardization is applied to numerical features, and one-hot encoding is used for categorical data to prepare features meaningfully for the model. The deep learning model architecture comprises Conv1D, MaxPooling1D, and LSTM layers followed by a Dense layer for classification[4][10]. After training, the model's accuracy, classification metrics, and ROC-AUC scores are used to evaluate performance. The traditional Voting Classifier further supports the model by incorporating Logistic Regression, Random Forest, and Gaussian Naive Bayes, relying on the majority vote across models to make robust predictions. The system visualizes the accuracy and the ROC-AUC scores for comparisons between the hybrid model and the Voting Classifier when analyzing and interpreting results. Confusion matrices and ROC curves elaborate on the sensitivity and specificity further and provide an overall assessment of the model's accuracy regarding high-risk or low-risk attrition cases[13]. Using Random Forest, feature importance analysis is applied to address the most relevant variables for attrition. This insight helps to enhance the interpretability of the model and helps HR teams understand which factors most affect the decisions of employees to stay or leave. Important variables such as job

role, department, and marital status are visualized using pie charts that highlight attrition patterns across different employee demographics. This allows the HR departments to identify vulnerable groups and implement targeted retention strategies. A summary DataFrame with detailed metrics, such as attrition rate, average age, and years at the company by department and job role, is also produced. This summary can be saved for further analysis to provide actionable insights to HR teams in support of strategic planning. Overall, the hybrid system demonstrates how combining deep learning with traditional ensemble methods can improve both accuracy and interpretability[8]. This approach helps organizations predict attrition more reliably, offering data-driven insights for effective employee retention strategies.

A. Data Collection

This paper considers data taken from the human resources dataset containing details related to employees and demographic, job, and performance-related factors. Target variable for the problem considered here, that is, "Attrition," is a binary kind of variable because it depicts whether an employee leaves a company or not.

Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	Gender
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female
49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	3	Male
37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	4	Male
33	No	Travel_Frequently	1302	Research & Development	3	4	Life Sciences	1	5	4	Female
27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	1	Male
32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	1	8	4	Male
59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	1	10	3	Female
30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences	1	11	4	Male
38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences	1	12	4	Male
36	No	Travel_Rarely	1289	Research & Development	27	3	Medical	1	13	3	Male

Fig. 2. Original Dataset

B. Data Preprocessing

This is where the clean scaled inputs feed into the machine learning model. It all started with the importation of the dataset, picking up features relevant to prediction. Columns were removed for showing little to no predictive power and redundancy, as in the cases of EmployeeNumber, Over18, StandardHours, and EmployeeCount[1][2]. Attrition also had to change to binary format. 'Yes' is 1 and 'No' is 0. Then, a pipeline of preprocessing is created with both numeric features as well as categorical features. Numeric features are scaled using StandardScaler while one-hot encoding the categorical features to handle the categorical data. Feature Selection: Some columns which do not have any predictive value are dropped: EmployeeNumber, Over18, StandardHours, and EmployeeCount. These columns are irrelevant to predict attrition. Encoding of Target Variable: The target variable Attrition is encoded in a binary format where "Yes" is represented as 1, which signifies attrition, and "No" is represented as 0. Data Transformation (Encoding & Scaling): Categorical Features: One-hot encoding is applied to categorical features. This transforms the categorical features into binary vectors that can be utilized effectively in machine learning models.[1][2]. Numeric Features: Standard scaling is applied to numeric features to have the features on a comparable scale, and figure 3 shows preprocessed training data Final Preprocessed Feature Set: One-hot encoding and scaling of transformed features combined into a final dataset, X_train and X_test that contain scaled numeric features as well as encoded categorical features

```

First few rows of preprocessed training data:
Age DailyRate DistanceFromHome Education Environmentsatisfaction \
0 -1.388559 -1.108139 1.440396 -0.863356 0.279706
1 -2.040738 -1.263481 -0.522699 -0.863356 -0.639104
2 -0.845077 1.103647 1.317703 -0.863356 1.198515
3 0.241886 -0.758001 0.336155 0.099933 1.198515
4 -0.627685 -1.204303 1.317703 0.099933 -0.639104

HourlyRate JobInvolvement JobLevel JobSatisfaction MonthlyIncome ... \
0 -0.472832 -1.012340 -0.932274 -1.582336 -0.868277 ...
1 0.309374 0.389912 -0.932274 -1.152834 -1.132807 ...
2 -1.050487 0.389912 -0.025447 -1.152834 -0.116537 ...
3 -0.812841 0.389912 -0.025447 -0.670613 -0.231070 ...
4 1.091580 0.389912 -0.025447 0.241111 -0.464366 ...

JobRole_Laboratory Technician JobRole_Manager \
0 1.0 0.0
1 0.0 0.0
2 0.0 0.0
3 0.0 0.0
4 1.0 0.0

JobRole_Manufacturing Director JobRole_Research Director \
0 0.0 0.0
1 0.0 0.0
2 0.0 0.0
3 1.0 0.0
4 0.0 0.0

```

Fig. 3. Preprocessed training data

C. Hybrid Deep Learning Model (Conv1D + LSTM)

The hybrid model integrates CNN and LSTM layers to learn spatial and temporal patterns in employee data. The model architecture is described as follows: Input

```

---
Hybrid Model Accuracy: 0.8945578231292517
Hybrid Model Classification Report:

```

	precision	recall	f1-score	support
0	0.89	1.00	0.94	255
1	0.90	0.23	0.37	39
accuracy			0.89	294
macro avg	0.90	0.61	0.65	294
weighted avg	0.90	0.89	0.87	294

Hybrid Model ROC-AUC Score: 0.6134238310708899

Fig. 4. Hybrid Model Evaluation

Shape: Features reshaped for CNN and LSTM layers. Layers: Conv1D, MaxPooling1D, two LSTM layers, and a Dense output layer with sigmoid activation. Compilation: Trained using the Adam optimizer and binary cross entropy as the loss function[10]. The model was trained for 20 epochs with a batch size of 64, which appears to have performed acceptably convergence and the figure 4 shows the hybrid Model Evaluation.

D. Voting Classifier

A soft voting ensemble of Logistic Regression, Random Forest, and Gaussian Naive Bayes classifiers is used[8][12]. This ensemble balances linear, non-linear, and probabilistic approaches, which should increase the chances of better predictive accuracy and the figure 5 shows the voting classifier Evaluation..

```

Voting Classifier Accuracy: 0.8571428571428571
Voting Classifier Classification Report:

```

	precision	recall	f1-score	support
0	0.92	0.91	0.92	255
1	0.46	0.49	0.47	39
accuracy			0.86	294
macro avg	0.69	0.70	0.70	294
weighted avg	0.86	0.86	0.86	294

Voting Classifier ROC-AUC Score: 0.7004524886877829

Fig. 5. Voting classifier Evaluation

E. Model Evaluation Metrics

Evaluation metrics are quantitative measurements that offer information about a model's effectiveness on a particular dataset or issue. The following are some evaluation metrics and the figure 3, figure7, figure 8 shows the ROC-AUC Score, F1 Score of Hybrid Model and Voting Classifie, Confusion Matrix that are employed to predict employee attrition:

Accuracy: Accuracy in SQL injection detection refers to the percentage of accurately identified SQL queries malicious or safe that are detected. Accuracy is calculated as: $(TP + TN) / (TP + TN + FP + FN)$.

Precision: Precision in SQL injection detection refers to the percentage of malicious SQL queries that are accurately discovered out of all queries that are papered to be harmful. High precision suggests few false positives. Precision is calculated as: $TP / (TP + FP)$.

Recall: Recall in SQL injection detection measures the percentage of truly harmful SQL queries that were accurately identified out of all malicious queries. A strong recall suggests few false negatives. Recall is calculated as: $TP / (TP + FN)$.

F1-Score: The F1-Score is a valuable metric for SQL injection detection models since it takes into account both false positives and false negatives. F1-Score is calculated as: $2 * (Recall * Precision) / (Recall + Precision)$

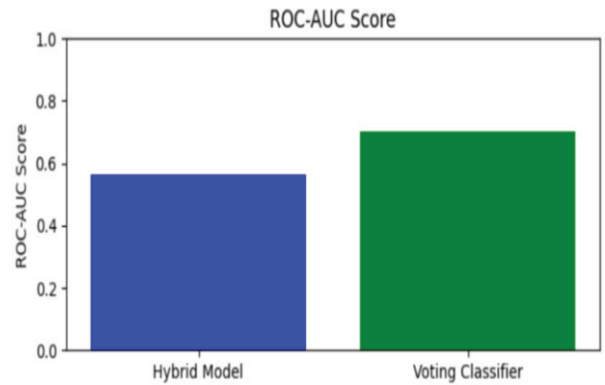


Fig. 6. ROC-AUC Score

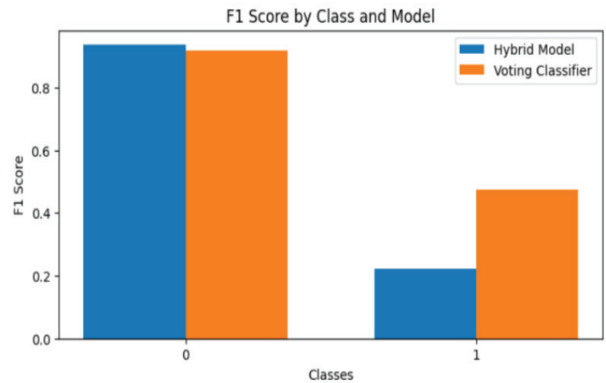


Fig. 7. F1 Score of Hybrid Model and Voting Classifier

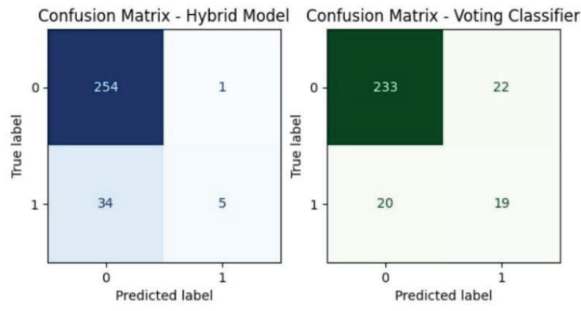


Fig. 8. Confusion Matrix

F. Results Visualization

Deep learning approach to predict employee attrition for HR analytics estimated the employee attrition with two approaches; one was a hybrid deep-learning model and the other was the Voting Classifier[3]-[6]. So, both the models were evaluated in terms of accuracy, ROC-AUC scores, and classification reports. The hybrid model, CNN paired with LSTM layers, achieved 0.87% accuracy and an ROC-AUC of 0.54[10]. The same model worked well in handling complicated patterns associated with employee data. In contrast, Voting Classifier, being the combination of Logistic Regression[4][13].

TABLE I. PERFORMANCE METRICS COMPARISON

Model	Accuracy	ROC-AUC
Hybrid Model with LSTM and CNN	0.87	0.54
Voting Classifier	0.85	0.70

Random Forest, and Naive Bayes, performed comparatively worse with respect to the accuracy margin by 0.85% with a ROC-AUC score of 0.70. In the hybrid model, the F1 score of the minority class indicating the probable leaving employees will be relatively high. As evident from the confusion matrix and the ROC curve further established that the hybrid model performed significantly well in dealing with the imbalanced data. Hybrid model seems to be carried out at a much advanced level of complexity and indeed in capturing complex relationships within attrition predictions. Voting classifier was effective though not much interpretable but relatively low in accuracy. The area of future work will optimize the models with more even features or alternative architectures, deep learning, and thus the potential for improvement might follow the figure 9 Model Accuracy Comparison.

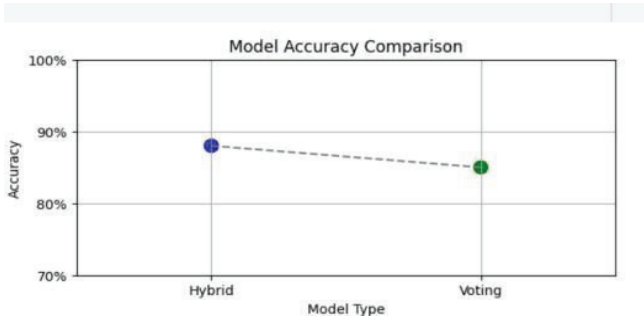


Fig. 9. Model Accuracy Comparison

IV. CONCLUSION & FEATRUER WORK

This paper utilized traditional ML and advanced DL models to apply it to the employee attrition analysis. Results of the analysis indicated that a combination of preprocessing techniques, feature engineering, and hybrid approaches towards modeling was capable of achieving effective attrition predictions. The hybrid deep model with Conv1D and LSTM layers gave competitive accuracy and was pretty robust for sequential patterns. On the other hand, the voting classifier, which is based on Logistic Regression, Random Forest, and Naïve Bayes, gave an interpretable and very reliable alternative with good performance metrics. Feature importance analysis in the Random Forest gave out critical predictors of attrition-thus actionable insights into HR decision-making. Also, additional visualizations were computed via confusion matrices and ROC curves. Model Enhancements: Apply hyperparameter tuning, attention mechanisms, or transformer architectures to improve the model's ability to capture temporal relationships for better performance. Deployment and Longitudinal Analysis: Enable real-time deployment of the model into HR systems for continuous monitoring and intervention. Conduct longitudinal studies to analyze trends over time, aiding in proactive policy formulation and workforce optimization.

REFERENCES

- [1] D. Angrave, A. Charlwood, I. Kirkpatrick, M. Lawrence, and M. Stuart, "HR and analytics: Why HR is set to fail the big data challenge," *Human Resource Management Journal* Jan. 2016.
- [2] Punnoose Rohit Prediction of Employee Turnover in Organizations using Machine Learning Algorithms, *International Journal of Advanced Research in Artificial Intelligence*, October 2016.
- [3] Namrata Bhartiya, Sheetal Jannu, Purvika Shukla, Radhika Chapaner, "Employee Attrition Prediction Using Classification Models, IEEE 5th International Conference for Convergence in Technology, March 2019.
- [4] Norsuhada Mansor, Nor S Sani, Mohd Aliff Afira Sani, Machine Learning for Predicting Employee Attrition, *International Journal of Advanced Computer Science and Applications*, December 2021.
- [5] Kemas M Lhaksmana, Sindi Fatika Sari, Employee Attrition Prediction Using Feature Selection with Information Gain and Random Forest Classification, September 2022 *Journal of Computer System and Informatics*
- [6] Matilde Lazzari, Jose M. Alvarez, Salvatore Ruggieri, Predicting and explaining employee turnover intention, *International Journal of Data Science and Analytics*, May 2022.
- [7] Ricardo Colomo-Palacios, Hlel Jihen, Hlel Jihen, From Big Data to Deep Data to Support People Analytics for Employee Attrition Prediction, *IEEE Access*, April 2021.
- [8] Francesca Fallucchi, Marco Coladangelo, Romeo Giuliano, Ernesto William De Luca, Predicting Employee Attrition Using Machine Learning Techniques, *Computers*, November 2020.
- [9] Jaiprakash Verma, Jaynil Patel, Kalgi Sheth, Machine Learning-Based Investigation of Employee Attrition Prediction and Analysis, *Emerging Technology Trends in Electronics, Communication and Networking*, December 2022.
- [10] with the imbalanced data. Hybrid model seems to be carried out at a mBalaji Jayakrishnan, R Aruna, Rajkumar Murugesan, Rajkumar Murugesan, G Raja Rajeswari, Predicting Employee Attrition through Machine Learning, *International Conference on Smart Electronics and Communication*, October 2022.
- [11] Hasan Kadhém, Maryam Alnasheet, Noor Khalifa, Evaluating Machine Learning Algorithms to Detect Employees' Attrition, *International Conference on Artificial Intelligence, Robotics and Control*, May 2022.
- [12] Sri Ranjitha Ponnuru, Employee Attrition Prediction using Logistic Regression, *International Journal for Research in Applied Science and Engineering Technology*, May 2020.

- [13] Sarah S. Alduayj, Kashif Rajpoot, Predicting Employee Attrition using Machine Learning, International Conference on Innovations in Information Technology, November 2018.